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2,436,881

DRILL CHUCK

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Fig. 1.

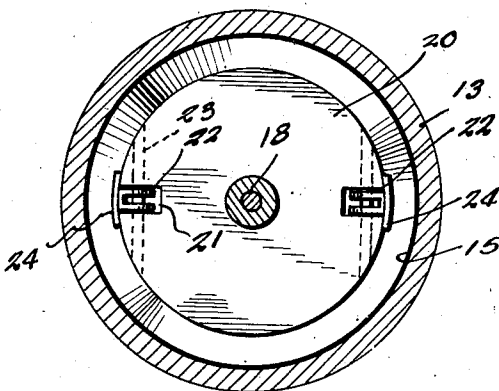
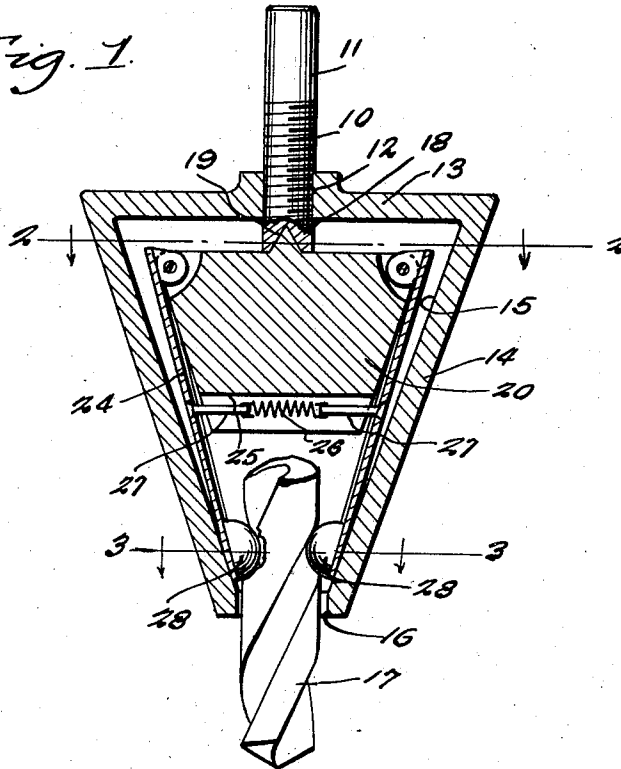
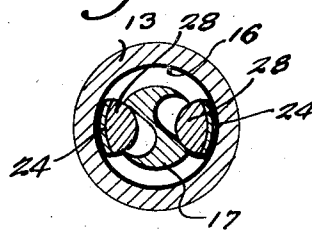


Fig. 2.

Fig. 3.



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DRILL CHUCK

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1 Claim. (Cl. 279—39)

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This invention relates to a drill chuck, and more particularly to such a chuck adapted for the holding of drills in drill presses or the like.

A primary object of this invention is to hold broken drills in related assembly with drill chucks, in such manner that they may be readily utilized, even though the shank of the drill has been broken, which breakage normally renders the drill inoperative.

An important feature of this invention is the provision of a special chuck adapted to be seated in a drill press or the like, and having means associated therewith for engaging the flutes or channels in a broken drill, in such manner that the same may be utilized for drilling operation even though its shank may be broken off.

An additional feature of the invention is the provision of such a device characterized by a frusto-conical member positioned for movement within a second frusto-conical member, the first member having extended pivoted arms provided with protuberances adapted to engage in the flutes of a broken drill.

A further feature of this invention is the provision of such a device which will be sturdy and durable in construction, reliable and efficient in operation, and relatively simple and inexpensive to manufacture, assemble and utilize.

Other objects will in part be obvious, and in part be pointed out as the description of the invention proceeds and shown in the accompanying drawing wherein there is disclosed a preferred embodiment of this inventive concept.

In the drawing:

Figure 1 is a longitudinal sectional view taken through one form of the chuck embodying this inventive concept.

Figure 2 is a sectional view taken substantially along the line 2—2 of Figure 1 as viewed in the direction indicated by the arrows.

Figure 3 is a sectional view taken substantially along the line 3—3 of Figure 1, as viewed in the direction indicated by the arrows.

Similar reference characters refer to similar parts throughout the several views of the drawing.

Having reference now to the drawing, there is shown at 10 a threaded member having a squared shank 11 adapted to engage in a drill press or similar device, the threads of which engage in a threaded aperture 12 in the base 13 of a frusto-conical member 14 having a hollow interior 15, and terminating at its smaller end in an aperture 16, through which a broken drill 17 is adapted to be inserted into the interior there-

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of. The lower end of the threaded member 12 is provided with a conical depression 18 adapted to engage a conical projection 19 carried on the base of a second frusto-conical member 20 positioned in the recess 15. The member 20 is provided with oppositely disposed cut-away portions 21 within which are pivotally mounted lugs 22 as on pivot pins 23, which lugs comprise inwardly turned ears of arms 24 which extend in opposite aligned relation substantially the full length of the recess 15, it being noted that the member 20 is materially shorter than the longitudinal extent of the recess.

A transverse channel 25 extends across the narrow portion of the member 20 between the arms 24, and contains a tension spring 26 connected to the inner ends of links 27 secured to the arms 24. The lower end of each arm 27 is provided with a hemispherical protuberance 28, the protuberances being oppositely positioned, and adapted to engage in the flutes or channels of the drill 17.

From the foregoing the operation of the device should now be readily understandable. When it is desired to utilize a broken drill 17, the shank 11 of the chuck is first inserted in the drill press or the like, and the drill passed through the opening 16 between the protuberances 28. Rotation of the member 13, now causes the same to move upwardly along the threaded portion 10, whereupon the recess 18 engages the conical protuberance 19 on the base of the member 20 forcing the same downwardly within the recess 15 in the interior of member 13. Such movement compresses the arms 24 in their associated spherical members 28, which engage in the flutes of the drill 17, the spring 26 assisting in the operation, and holding the drill lightly until such time as the members are firmly seated by the interior configuration of the recess, whereupon when the assembly is tightened the broken drill is held securely in position, and in related assembly with the drill chuck whereby it may be utilized for drilling operations.

It will thus be seen that there is herein provided an improved device accomplishing all the objects of this invention and others, including many advantages of great practical utility and commercial importance.

As many embodiments may be made of this inventive concept and as many modifications may be made in the embodiment hereinbefore shown and described, it is to be understood that all matter herein is to be interpreted merely as illustrative and not in a limiting sense.

I claim:

A drill supporting chuck comprising a hollow conical shaped outer member closed at its large end and open at its small end, said closed end being formed with a threaded centrally disposed opening, a threaded shank adjustably disposed in said threaded opening, a solid frusto-conical member supported by said shank disposed in said hollow outer member for movement lengthwise thereof, said solid frusto-conical member being formed with oppositely disposed bearing slots in its upper end, pivot pins extending transversely of said slots, arcuate depending clamping arms having spaced apertured ears on their upper ends pivotally supported upon said pivot pins and having inwardly extending locking protuberances on their lower ends adapted to engage in the flutes or channels of a drill when inserted within the open end of said outer member, said inner member being formed with a transverse channel in its lower end, inwardly extending arms secured to said clamping arms disposed in said transverse

channel, and a coil spring secured between the inner ends of said inwardly extending arms for resiliently urging said clamping arms and protuberances inwardly to positively lock and support a drill when inserted in a chuck.

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